

TD 4

Exercice 1 – Support Vector Machine

We consider here a binary classification problem with $Y = \{-1, +1\}$ for data in a description space $X \in \mathbb{R}^d$. Let $\{(\mathbf{x}^i, y^i) \in (X, Y)\}, i \in \{1, \dots, n\}$ be the training set. The decision function of the classifier is given by : $f_{\mathbf{w}, b}(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x} + b)$.

First, we consider a linearly separable dataset. This dataset and the decision boundary are represented (in 2D) in Figure 1.

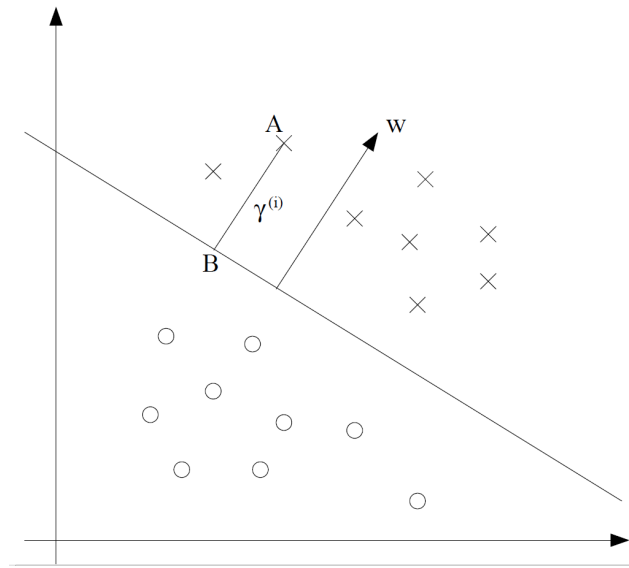


FIGURE 1 – Linearly separable dataset

Q 1.1 Margin

In this figure, the sample $\mathbf{x}^i$ with label y^i is represented by point A. We are interested in its signed distance γ^i to the decision boundary (where the closest point is represented by B in the figure).

Q 1.1.1 Given that $\frac{\mathbf{w}}{\|\mathbf{w}\|}$ is a unit vector orthogonal to the decision boundary, express γ^i in terms of $\mathbf{x}^i$, y^i , $\mathbf{w}$, and b .

Q 1.1.2 Show that the distance and the solution do not change when multiplying the solution by a scalar, i.e., for $(\alpha \mathbf{w}, \alpha b)$. What does this imply if we want to maximize the geometric distance from the decision boundary?

Q 1.2 SVM Formulation

We now consider the following constrained optimization problem :

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s.t.} \quad & y^i (\mathbf{w}^T \mathbf{x}^i + b) \geq 1, \forall i \in \{1, \dots, n\} \end{aligned}$$

Q 1.2.1 Why do we choose the constraint ≥ 1 instead of ≥ 0 ? Why specifically 1?

Q 1.2.2 Set up the Lagrangian to optimize this problem under constraints.

Q 1.2.3 Give the analytical solution for minimizing this Lagrangian with respect to $\mathbf{w}$ and b .

Q 1.2.4 Deduce a new "dual" formulation of our optimization problem under constraints.

Q 1.2.5 What does this new formulation allow ?

Q 1.2.6 What is the issue with the optimization problem we considered ? Propose a new formulation to correct this issue.

Q 1.2.7 What happens if $\xi_i > 0$ for some i ?

Exercise 2 – Kernels

Q 2.1 Show that if K and K' are two kernels (i.e., there exist ϕ and ϕ' such that $K(x, y) = \langle \phi(x), \phi(y) \rangle$, $K'(x, y) = \langle \phi'(x), \phi'(y) \rangle$) :

Q 2.1.1 cK is a kernel for $c \in \mathbb{R}^+$;

Q 2.1.2 $K + K'$ is a kernel ;

Q 2.1.3 KK' is a kernel ;

Q 2.1.4 $(1 + \langle x, x' \rangle)^d$ is a kernel.

Exercise 3 – Kernels on String Sequences

Let S be a sequence of words over a finite alphabet $\mathcal{A}$. Show that :

1. $K(x, x') =$ the number of common substrings of length 5 between x and x' is a kernel ;
2. $K(x, x') = 1$ if x and x' have at least one common substring of length 5, and 0 otherwise, is not a kernel (hint : consider 3 strings x , x' , and x'').